



Q. Water hammer pulse is seen in which of the following conditions?

- a. Aortic stenosis
- b. Aortic regurgitation
- c. Mitral stenosis
- d. Mitral regurgitation

Ans: (b)

Q. Pistol shot sounds over femoral artery on auscultation in a patient with a history of rheumatic fever, a diastolic murmur, and a dilated left ventricle is called as?

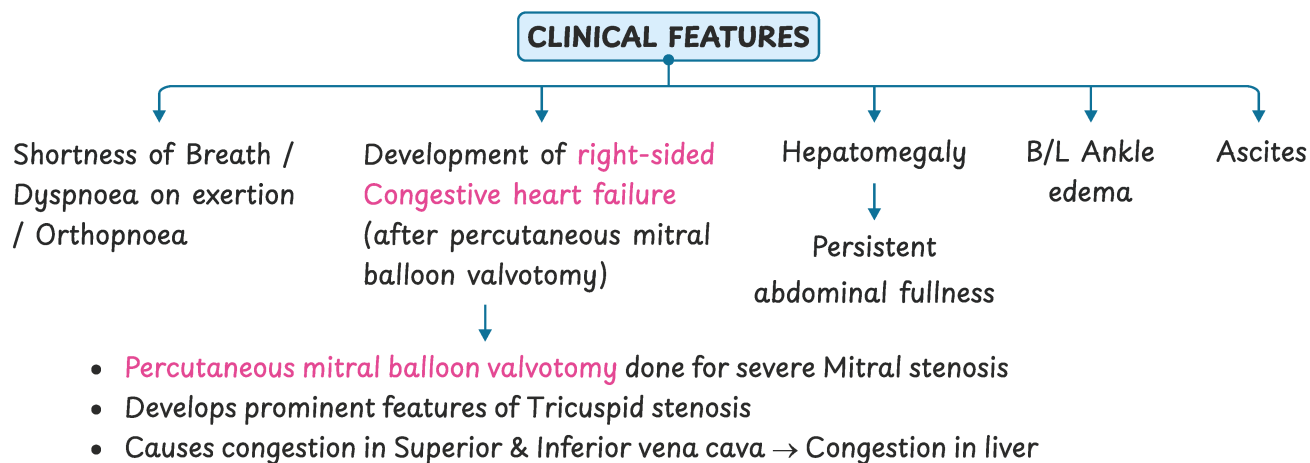
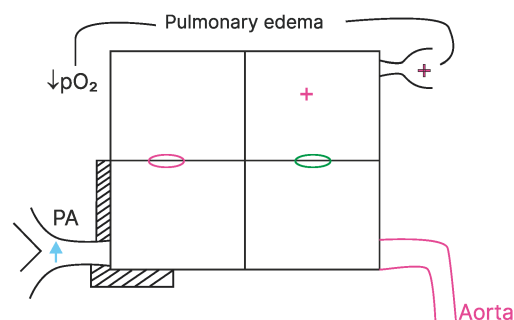
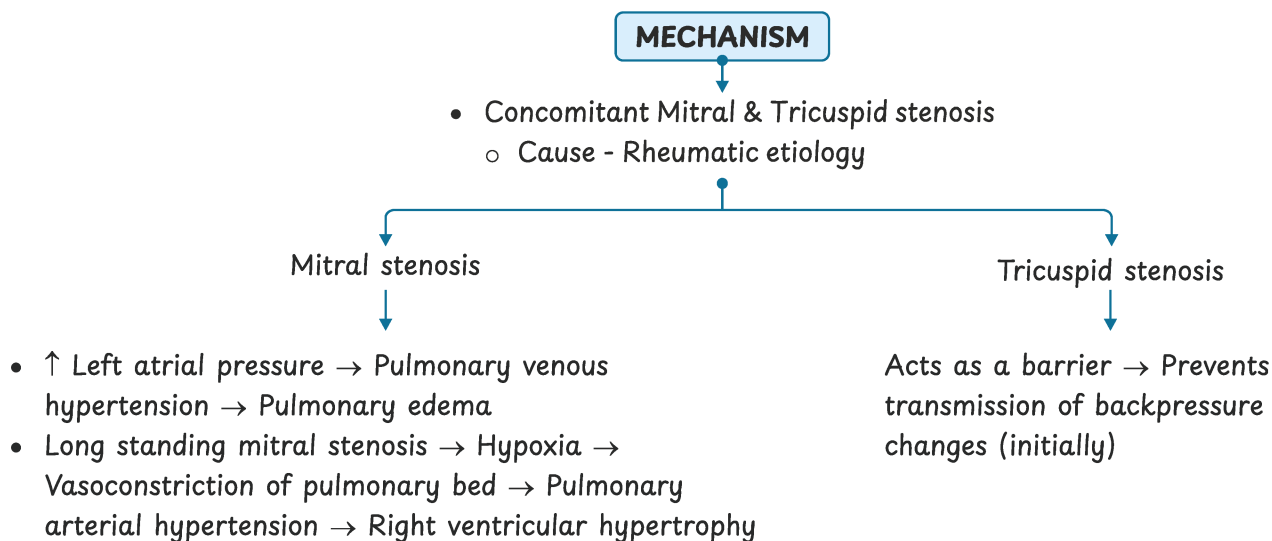
- a. Traube's sign
- b. De Musset's sign
- c. Hill's sign
- d. Duroziez sign

Ans: (a)



3. TRICUSPID STENOSIS

TRICUSPID STENOSIS MECHANISM

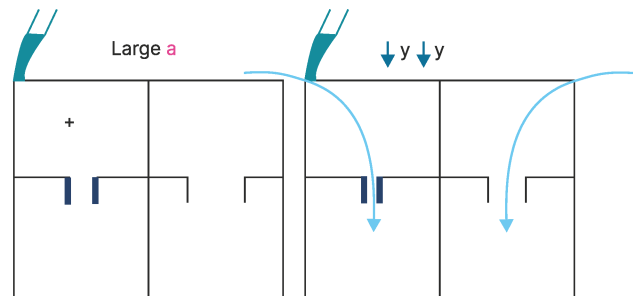


1 or 2 'a' waves

- ↑ pressure in Right atria → Congestion → ↑ force required to pump blood on contraction
- On JVP examination (in internal/external jugular vein)
 - Column of blood goes much higher

Blunted 'y' descent

- Obstruction (Tricuspid stenosis) → ↓ ventricular filling (on relaxation)



Pulsatile liver (Presystolic pulsations)

- Tricuspid stenosis operates in diastole

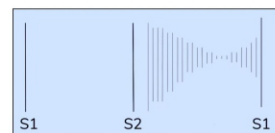
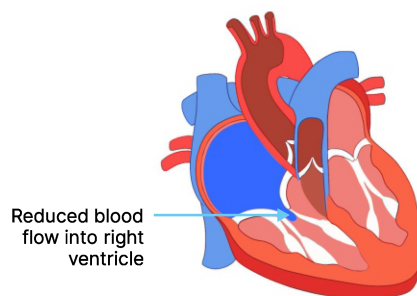
Mid diastolic murmur

- ↑ on inspiration
- Heard in **left lower sternal border** (Tricuspid area)

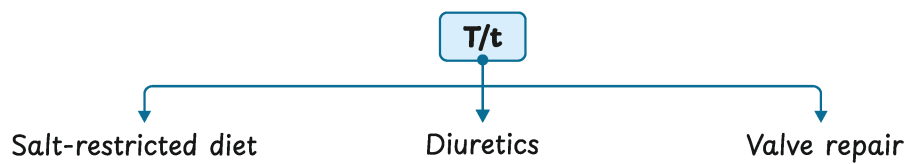
WORK-UP

00:12:30

ECG	ECHO	CHEST X RAY	ULTRASOUND
• Right atrial enlargement - P-pulmonale ○ Vertical height of P-wave → In Limb leads > 2.5 mm → In Chest leads > 1.5 mm • Right Ventricular Hypertrophy	• IOC • Right ventricular enlargement ○ Complication of Mitral stenosis ○ If no Right ventricular enlargement - Isolated Tricuspid stenosis • Severe tricuspid stenosis - Orifice size < 1 cm² • Transit time > 190 msec ○ Time taken for blood to reach Right ventricle from Right atria	• RV enlargement ○ Presence of pre-existing Mitral stenosis • Congestion in Right atria, Superior vena cava, Azygos vein	• Free fluid in peritoneal cavity • ↑ liver span



Mid Diastolic Murmur





- Q. Evaluation of a patient revealed the presence of mid diastolic murmur. JVP showed prominent a-waves. What is the likely diagnosis?
- a. Mitral stenosis
 - b. Tricuspid stenosis
 - c. Mitral regurgitation
 - d. Tricuspid regurgitation

Ans: (b)

- Q. A 60-year-old male presents to the clinic due to fatigue and leg swelling for the past few weeks. On examination, jugular venous distention, pulsatile hepatomegaly and mid diastolic murmur is heard. The patient's history is significant for PMBV done last year. What is the most likely underlying cause for this presentation?
- a. Tricuspid stenosis
 - b. Pulmonic stenosis
 - c. Aortic regurgitation
 - d. Mitral regurgitation

Ans: (a)



TRICUSPID REGURGITATION, PULMONIC STENOSIS AND PULMONIC REGURGITATION



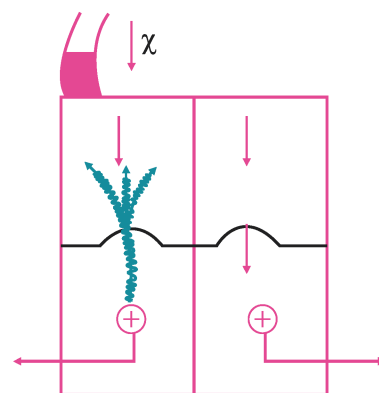
TRICUSPID REGURGITATION

- Leaking of blood from right ventricle into right atria → **Pansystolic murmur**

CAUSES

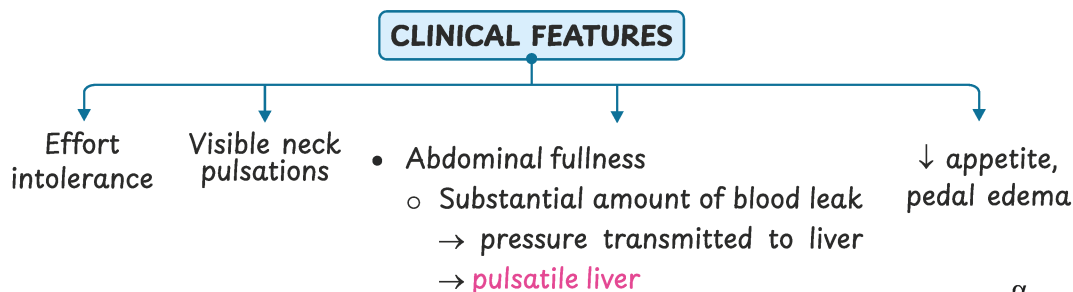
00:00:47

- Functional - **Annular** dilatation of Tricuspid valve
 - No damage to valve leaflets/Chordae tendinea
- R**heumatic, **R**adiation
- E**ndomyocardial fibroelastosis
- E**bstein anomaly
 - Intake of Lithium during pregnancy
 - Tricuspid stenosis/regurgitation
 - A/w** WPW syndrome
- P**ulmonary artery hypertension (**> 55 mmHg**) → Right ventricular failure
- P**acemaker - RV pacing wire → structural damage
- Carcinoid syndrome → **TIPS**
 - Right ventricular inflow tract - affected
 - T**ricuspid **i**nsufficiency
 - P**ulmonic **s**tenosis

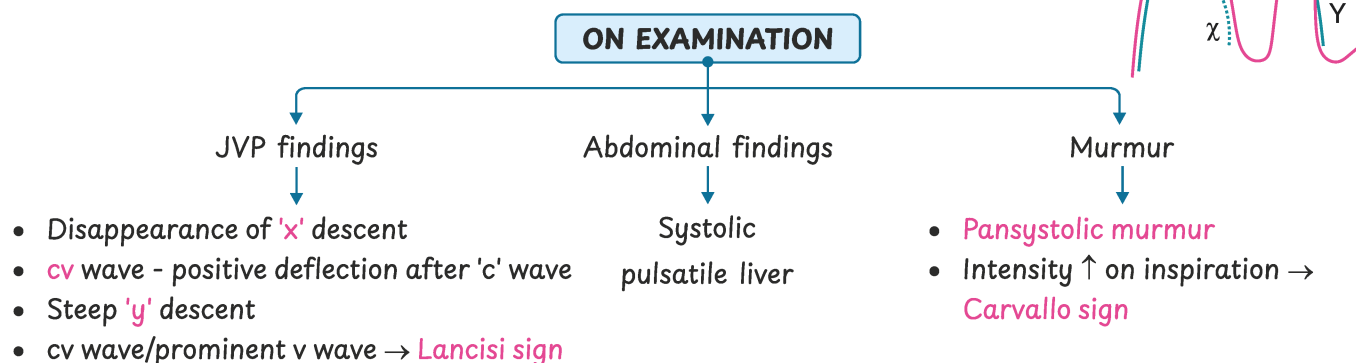


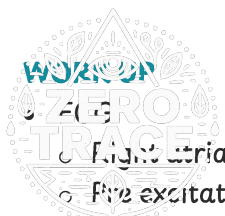
CLINICAL FEATURES

00:08:29



ON EXAMINATION

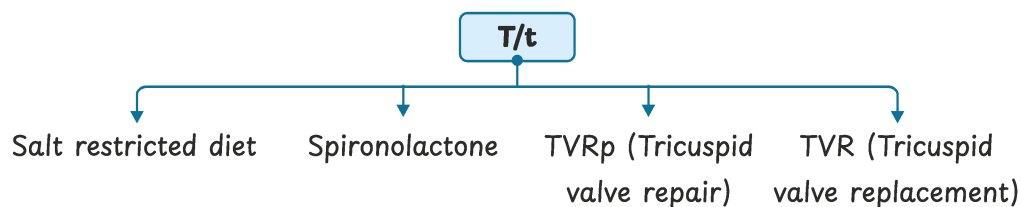




- Right atrial enlargement
- First excitation syndrome (Delta wave)
- Chest X-ray - RV enlargement
- Echocardiography - **IOC**
- Doppler - Reversed systolic flow in liver

T/t

00:19:11



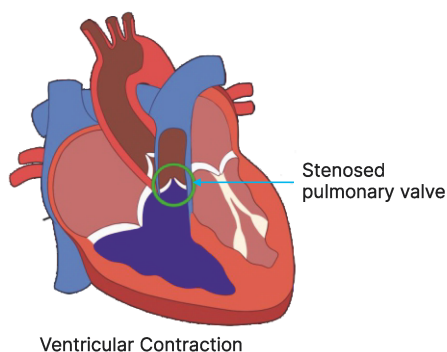
Important Information

- Dilated cardiomyopathy → Functional TR

PULMONIC STENOSIS

00:21:39

- Mnemonic - **PASS** (Pulmonic/ aortic stenosis → systolic murmur)
- Ejection systolic murmur



CAUSES

- Congenital
- Carcinoid syndrome - TIPS
- **Noonan** syndrome (45 XO)
 - Dysplastic valve
 - **PTPN1** gene defect on Chromosome 12
 - Webbed neck, Short stature
 - Pulmonic stenosis
- Rheumatic fever (±)

00:22:54



Important Information

- Turner syndrome (45XO)
 - Webbed neck, Short stature
 - Bicuspid aortic valve
 - Coarctation of Aorta

SEVERE PULMONARY STENOSIS

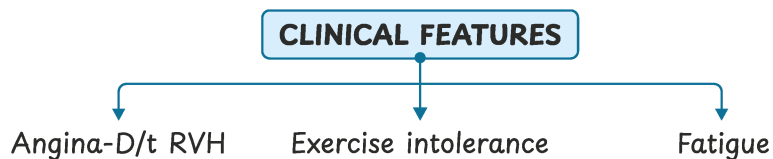
- Pressure gradient ≥ 50 mmHg
- **Ejection click** (↑ turbulence at mid systole)
- Forcible ejection **click** → Crescendo decrescendo murmur
- In ↑ pressure gradient, ejection click appears earlier
- All right sided events ↑ with inspiration **except** pulmonary ejection click



00:26:29

CLINICAL FEATURES

00:30:44

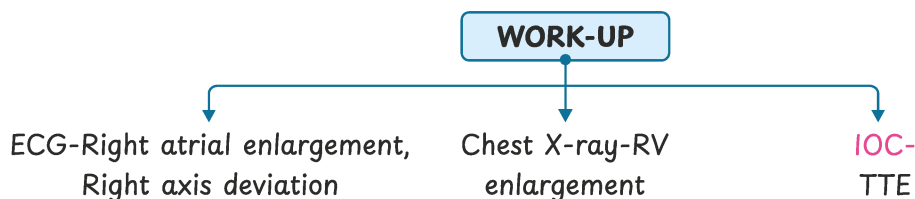


ON EXAMINATION

- P₂-soft → single S₂
- S₄ (+)-right sided 4th heart sound
- Murmur -**ejection systolic murmur**

WORK-UP

00:33:00

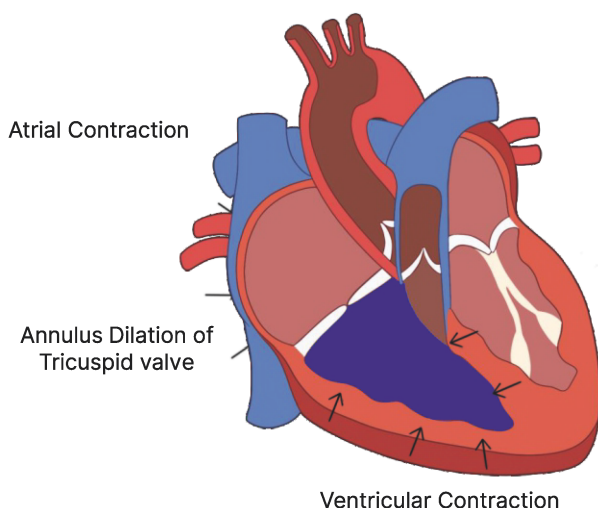


T/t

- Percutaneous pulmonic valvotomy

PULMONARY REGURGITATION

00:34:38



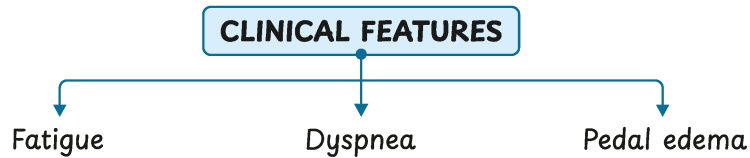
- **Early diastolic murmur (decrescendo murmur)**

CAUSES

- Primary artery hypertension
- Severe PAH - **annular dilatation** of pulmonic valve
- Tetralogy of Fallot repair
 - Over correction of sub-pulmonic stenosis
- Post balloon valvotomy
 - Over correction of pulmonic stenosis
- Marfan syndrome
 - **Fibrillin** gene defect on chromosome 15

00:36:39

CLINICAL FEATURES



ON EXAMINATION

- Murmur - Early diastolic/ Decrescendo murmur

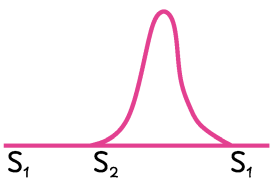
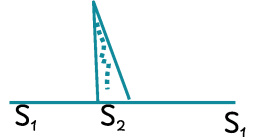
WORK -UP

00:40:25

- Cardiac MRI - **IOC**

T/t

- Transcatheter pulmonic valve replacement
- **Ross procedure**
 - Done for damaged aortic valve
 - Pulmonary autograft to replace diseased Aortic valve

	TS	TR	PS	PR
JVP	Large 'a' Blunted 'y'	Absent 'x' (+) cv wave Steep 'y'	Large 'a'	Large 'a'
MURMUR	Mid diastolic murmur 	Pansystolic murmur 	Ejection systolic murmur 	Early diastolic murmur 
PULSATION OF LIVER	Presystolic pulsation	Systolic pulsation	(-)	(-)



Q. A 32 year old male presents to the clinic due to palpitations, fatigue and RUQ discomfort due to hepatomegaly. Auscultation reveals a pansystolic murmur that is louder on inspiration. ECG shows right axis deviation. Echocardiography reveals defective coaptation of leaflets and there is leakage of blood from the right ventricle to the right atrium with reverse systolic flow in the liver on a Doppler. Which of the following JVP findings would be found in this patient?

- a. Large a wave
- b. Giant CV wave
- c. Blunted Y descent
- d. Steep x descent

Ans. (b)

Q. A known case of PAH presented with dyspnea associated with fatigue and pedal edema. Auscultation findings: early diastolic murmur over the left 2nd intercostal space. What is the most probable diagnosis at this stage?

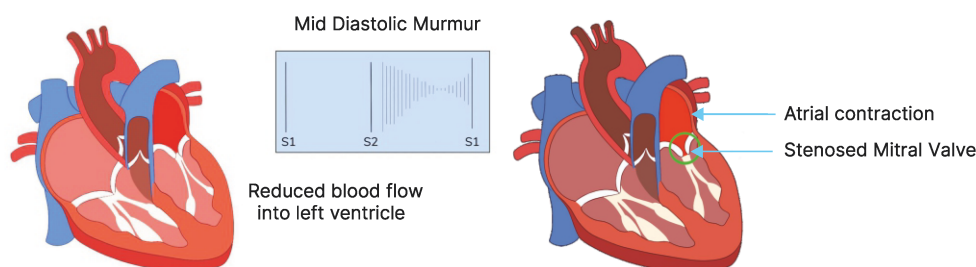
- a. Pulmonary regurgitation
- b. Aortic regurgitation
- c. Mitral stenosis
- d. Tricuspid stenosis

Ans. (a)

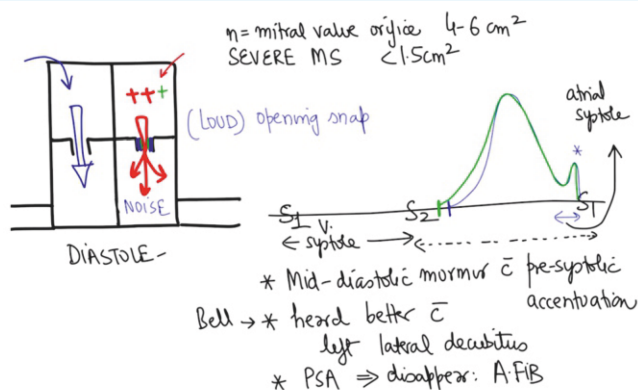
MITRAL STENOSIS (MS)

00:00:24

- mitral valve orifice
 - Normal : 4-6 sq. cm
 - Severe mitral valve stenosis: < 1.5 sq. cm
 - Very severe mitral valve stenosis: < 1 sq. cm
- Atria works against stenotic mitral valve → Dilated left atria → structural damage → triggers atrial fibrillation
 - Atrial fibrillation favors twitching of heart and formation of clots
 - Embolize to systemic circulation → embolic stroke



MID DIASTOLIC MURMUR	OPENING SNAP	PRE-SYSTOLIC ACCENTUATION (PSA)
- Cause: Turbulence - Caused by narrowing of mitral valve - Subsequently heard after opening snap	- Loud - Cause: Higher pressure in left atria	- Small secondary spike (with mid-diastolic murmur): - Low pitched murmur - Cause: Atrial contraction (systole) in stenosed valve - Heard better with bell → in left lateral decubitus position - If PSA disappears → Atrial fibrillation

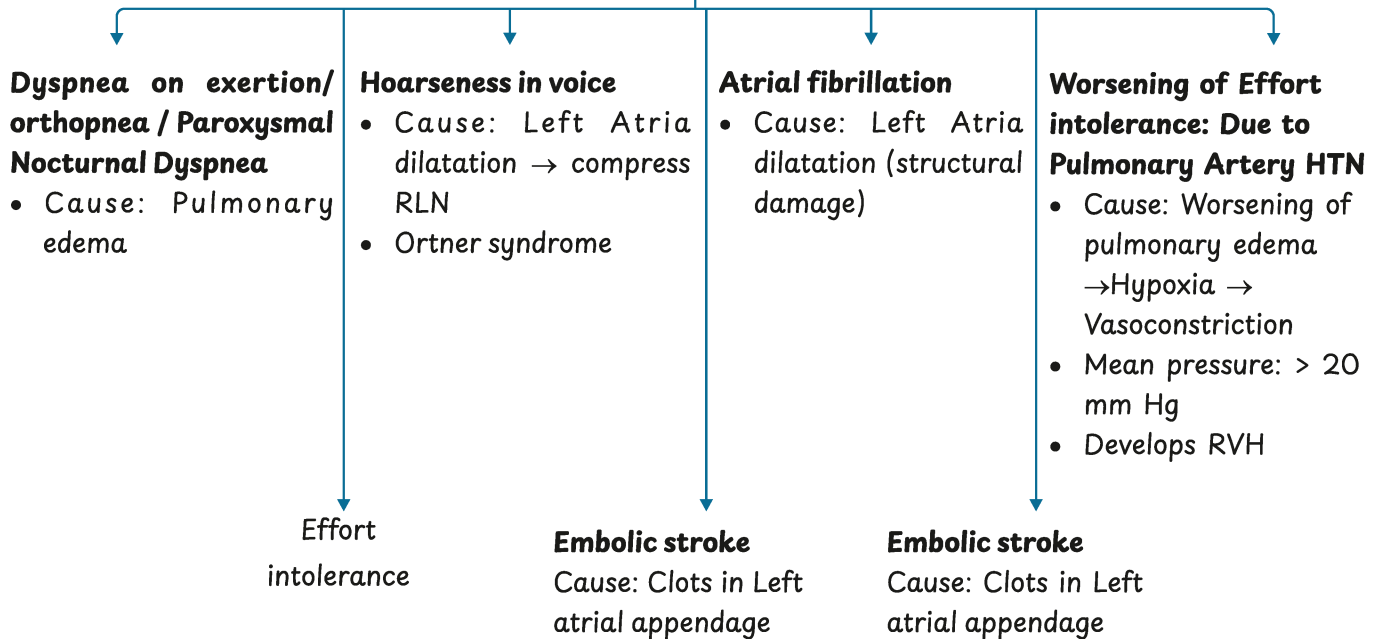


Important Information

- Severity of MS is determined by:
 - Length/duration of the murmur
 - Inversely proportional to S2 – opening snap gap.

CLINICAL FEATURES

00:15:02



Important Information

- T/t of Valvular Atrial Fibrillation: Warfarin (↓ risk of embolic stroke)
- T/t of Non-valvular atrial fibrillation: Apixaban (Novel oral anticoagulants) (↓ risk of embolic stroke)

ON EXAMINATION

00:25:58

→ Apex beat: Tapping

→ S1:

Types of S1	Causes
Loud	• ↑ transvalvular gradient ◦ Opening & closure of mitral valve with more force
Soft S1	• Calcified Mitral Stenosis: Elastic recoil ↓
Narrow split S1	• ↑ Transit time → Late closure of mitral valve → ↓ gap between M1 & T1
Single S1	• Worsening of Mitral stenosis → ↓↓↓ gap between M1 & T1 → M1 & T1 close at same point of time
Reverse split S1	• Worsening of mitral stenosis → ↓↓↓ gap between M1 & T1 → M1 closes after T1 closure



• S1 A2 P2: Normal

• In Pulmonary Artery Hypertension: **Loud P2**

→ S3: Absent

→ Low pitch murmur

- Mid Diastolic Murmur with Presystolic Attenuation heard

Important Information

- Posterior most chamber of heart: Left atria

WORK UP

00:31:10

ECG	CHEST X-RAY	TRANSTHORACIC ECHOCARDIOGRAPHY (TTE)
• Normal p wave ◦ Height: < 2.5 mm (limb leads) ◦ Duration: < 120 msec • In LA enlargement: ↑ duration of P wave (P-mitrale: >120 msec) • In Right Atrial enlargement (PAH): ↑ height of P wave (P-Pulmonale: >2.5mm)	• Straightening of left heart border ◦ Cause: Convexity of the pulmonary artery area (in PAH) & fullness left atrial appendage • Double atrial shadow ◦ Cause: Enlarged Left Atria pushing Right Atria • Batwing (pulmonary) edema • Widened carinal angle (Tracheal bifurcation) ◦ Normal: 60-70°	• IOC • Candle flame jet ◦ Cause: Turbulence • Hockey stick appearance ◦ Cause: Turbulence causing displacement of damaged mitral valves

Important Information

Hockey stick sign

- MRI Head
 - Variant Creutzfeldt-Jakob disease (Prion)
- ECG
 - Digoxin (ST depression)
- Echo
 - Mitral stenosis

**MANAGEMENT****Severe Mitral Stenosis**

- Percutaneous mitral balloon valvotomy (PMBV)

Mild to moderate Mitral Stenosis

- Diuretics – Spironolactone (K⁺ sparing)
- Digoxin

Calcified severe Mitral Stenosis

- Prosthetic heart valve

Bioprosthetic device

- ↓ Durability
- No requirement for oral anticoagulant Therapy
- Made of:
 - Homograft (from humans)
 - Xenograft (from pig, cows)

Metallic heart valve

- Durable
- Disadvantage: Lifelong oral anticoagulant Therapy

In Development of Pulmonary Artery Hypertension:

Mean Pulmonary Artery pressure > 20mm Hg

Loud P2 heard on auscultation

Vasoreactivity test

(inhaled NO): Done for diagnosis

Response

Positive test

T/t: Oral amlodipine

No response

Negative test

T/t: Bosentan + Epoprostenol + Sildenafil (Triple therapy)

Important Information

- **Percutaneous mitral balloon valvotomy (PMBV) – C/I:**
 - Mitral stenosis with Mitral regurgitation
→ Procedure causes worsening of Mitral regurgitation
 - Mitral stenosis with Left atrial appendage thrombus
 - Calcified severe Mitral Stenosis
→ Force of balloon → destroy calcified valve → Embolize
- Exhaled NO: used for evaluation of asthma



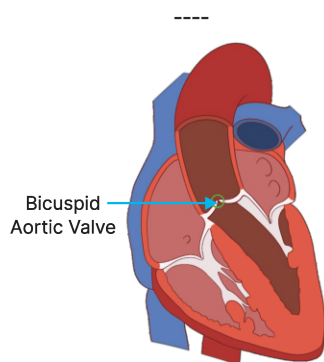
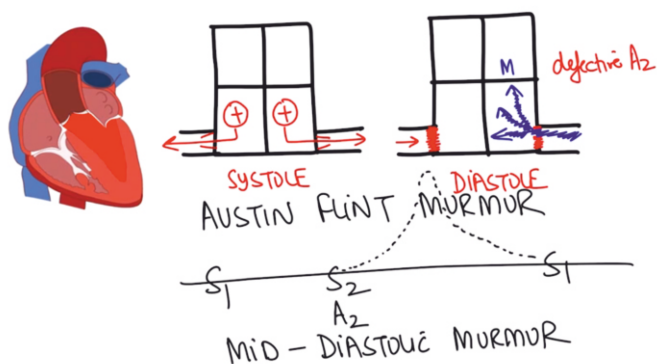
SEVERE AORTIC REGURGITATION (AR)

Severe Aortic Regurgitation

Severe defective aortic valve closure (defective A2 causing leakage of blood from aorta to left ventricle)

Regurgitant jets of blood hits mitral valve leaflet

Produces Austin Flint murmur
(Mid-diastolic Murmur)



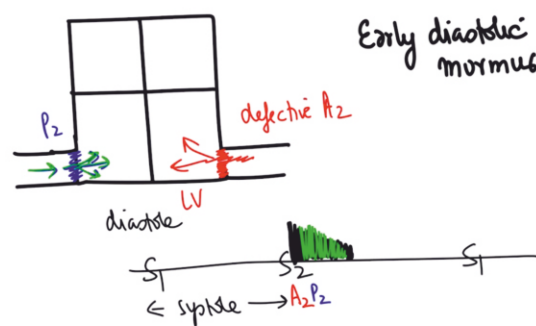
MILD AORTIC REGURGITATION

Mild Aortic Regurgitation

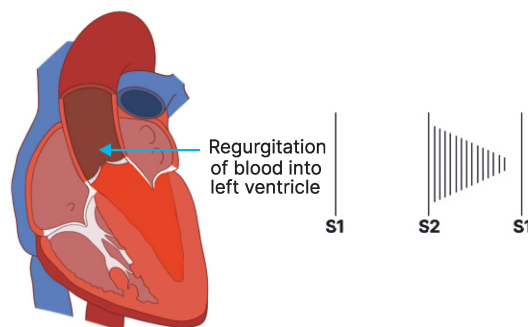
Mild defective aortic valve closure (defective A2) (causing minimal leakage of blood from aorta to left ventricle)

Minimal amount of Blood hits mitral valve leaflet

Produces Early diastolic murmur
(Decrescendo murmur / Seagull murmur)



- Murmur arises from S2 (A2) → dies down very soon (shorter duration)



MILD PULMONIC REGURGITATION

- Minimal blood leaking back to the right ventricle leading (defective P2) → shorter duration early diastolic murmur.
- Murmur arises from S2(P2)
- Murmur of Mild Aortic Regurgitation heard earlier compared to Murmur of Mild Pulmonic Regurgitation

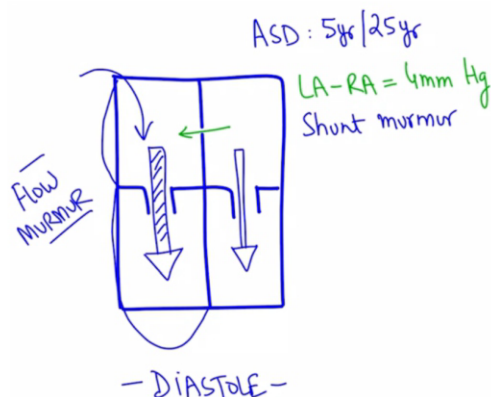


TYPES OF DIASTOLIC MURMUR

01:05:22

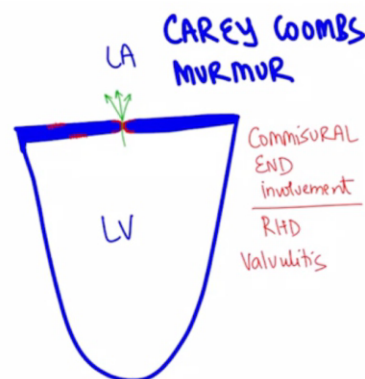
Early diastolic murmur (Mnemonic: G.A.P)	• Graham Steele murmur ◦ Cause: Severe Pulmonary Artery Hypertension → mild Pulmonic Regurgitation • Mild Aortic regurgitation • Mild Pulmonic regurgitation
Mid-diastolic murmur (Mnemonic: C.A.M - F)	• Carey Coombs murmur ◦ Cause: Rheumatic heart disease • Austin Flint murmur ◦ Cause: Severe Aortic regurgitation • Mitral stenosis • Flow murmur

FLOW MURMUR



- Right atrium receives blood from left atrium & venous return (In ASD - Down's syndrome)
 - Amount of blood received:
 - Right ventricle > Left ventricle
 - Pressure difference (Left atria - Right atria) = 4 mm Hg
- More blood pumped across tricuspid valve → Causes volume overloading of right side of heart → more turbulence produced
- Heard in diastole (maximum filling of blood)
- Shunt murmur: Absent
- Manifests at: 5 years of age
 - Rarely seen in 25 years of age

CAREY COOMBS MURMUR



- Aschoff nodules seen on superior & inferior surface of heart valves
 - Maximum concentration of Aschoff nodules → seen on **Commissural ends** (valve cusps touching)
 - **Valvulitis** (inflammation of valve cusps)
 - Causes leakage of blood from Left ventricle to left atria → left atria dilation
- During diastole → leaked blood in left atria leaks back to left ventricle
 - Mid-diastolic murmur
- Is a flow murmur
- Characteristic murmur of Rheumatic Heart Diseases

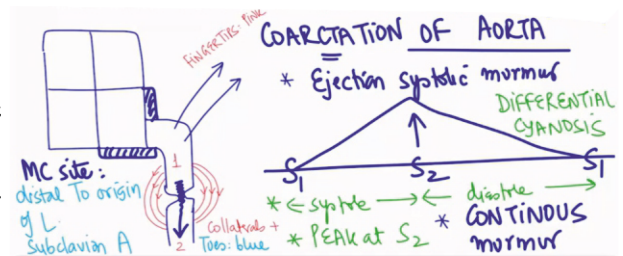
Important Information

- M/c murmur in children: Innocent murmur (systolic murmur)
- o Cause: Hyperdynamic circulation (due to fever)

COARCTATION OF AORTA

01:19:10

- Congenital heart defect
- Seen in Turner syndrome (45XO)
- M/c site:
 - Aortic isthmus (Distal to origin of left subclavian artery)
- On auscultation:
 - Ejection systolic murmur: initial
 - o Initial turbulent blood flow across aorta
 - Continuous murmur (In systole & diastole): later
 - o Peak at S2
- Clinical features:
 - Chest pain
 - o Cause: LVH → ↑ O₂ demand → hypoxia
 - Differential cyanosis:
 - o Fingertips - pink
 - o Toes: blue (↓ blood supply)
- Compensatory mechanism:
 - Development of collaterals (thin-walled vessels) originating from aorta
 - Blood flows across collaterals from high pressure proximal area to low pressure distal area
 - o Produce continuous murmur (In systole & diastole)



CONTINUOUS MURMURS

01:25:12

Coarctation of aorta

Patent ductus arteriosus

Mammary souffle:
↑ Internal Mammary Artery flow
• Physiological finding in pregnancy

Venous hum
• Cause: Dilated veins in chest

Rupture of Sinus Valsalva

• Causes leakage of blood from aorta (high pressure) to right atria (low pressure)

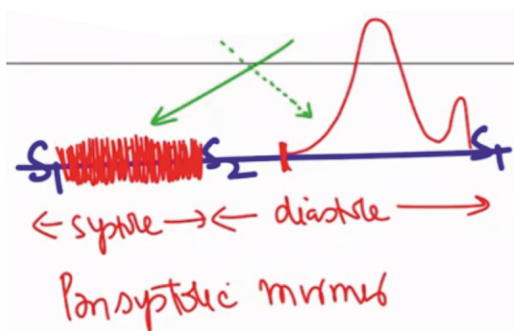
Peripheral pulmonic stenosis

• Narrowing of pulmonary arteries



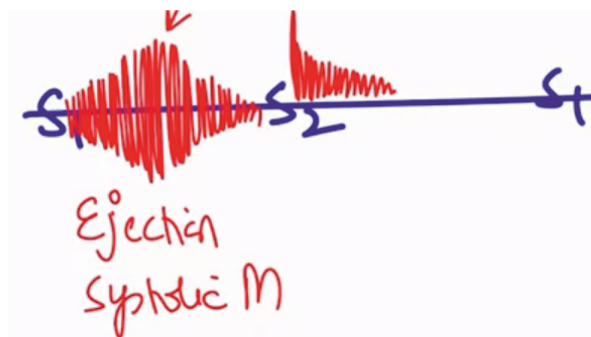
MITRAL STENOSIS WITH MITRAL REGURGITATION

- Pansystolic murmur: in Systole (MR)
- +
- Mid diastolic murmur (after opening snap): in Diastole (MS)
- Peak is not present at S2 → not a continuous murmur



ATRIAL STENOSIS WITH ATRIAL REGURGITATION

- Ejection systolic murmur: in Systole (AS)
- +
- Decrescendo murmur: in Diastole (AR)
- Peak is not present at S2 → not a continuous murmur



TYPES OF SYSTOLIC MURMUR

01:34:01

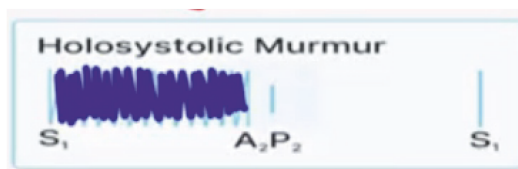
Ejection systolic murmur (Mnemonic: PASS)

- Pulmonic stenosis:
 - Carcinoid syndrome:
 - M/c valvular lesion: Tricuspid Insufficiency Pulmonic Stenosis (TIPS)
 - Noonan syndrome
- Aortic stenosis
- Hypertrophic Obstructive Cardiomyopathy (sub-valvular aortic stenosis)



Pansystolic murmur

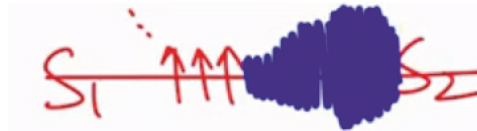
- Ventricular septal defect
- Chronic Mitral regurgitation
- Chronic Tricuspid regurgitation





Mid-systolic clicks f/b late systolic murmur

- Mitral Valve Prolapse / Barlow syndrome
 - **Cause of mid-systolic clicks:** Stretching of chordae tendineae
 - **Cause of late systolic murmur:** Leakage of blood



Early systolic murmur

- Acute Mitral Regurgitation

Important Information

- M/c Valvular lesion in Turner syndrome: Bicuspid aortic valve > Coarctation of aorta

	ALL MURMURS OF	
	Right side	Left side
On deep inspiration	• ↑	• ↓
On valsalva, standing, amyl nitrate (↓ venous return)	• ↓ ○ Exceptions: → Hypertrophic obstructive cardiomyopathy murmur: Louder → Mitral Valve Prolapse murmur: Longer	

HYPERTROPHIC OBSTRUCTIVE CARDIOMYOPATHY

01:44:17

Hypertrophied Intraventricular septum



Reduces cavity size of left ventricle



On standing, Venous return ↓



↓ blood in LV



Tunnel narrows

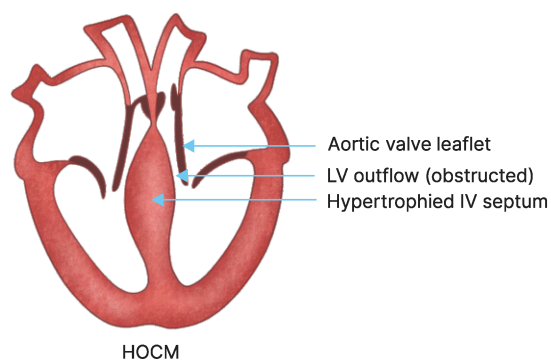
(Left ventricular outward tract obstruction)



↑ turbulence



Louder murmur





- Q. A 20-year-old woman presents with breathlessness & chest pain. She is a known case of mitral stenosis. Her pulse is irregularly irregular. Left atrial thrombus is seen on echocardiography. What is the best agent to prevent future thrombotic events?
- a. Dabigatran
 - b. Apixaban
 - c. Oral warfarin
 - d. Aspirin + Clopidogrel

Ans (c)

- Q. A male patient had feeble femoral pulses with upper limb blood pressure of 186/90 mm Hg. Chest X-ray showed enlarged intercostal arteries. What is the likely diagnosis?
- a. Coarctation of aorta
 - b. Atrial septal defect
 - c. Bicuspid aortic valve
 - d. Patent ductus arteriosus

Ans (a)